ICAR-KVK SOIL TESTING FACILITY, BELAGAVI

SOIL HEALTH & FERTILITY TEST CERTIFICATE

SAMPLE DATA — FOR PROJECT DEMONSTRATION ONLY (FICTIONAL TEST DATA)

Soil Sample ID:	DEMO-SHC-KA03	**Report Date:**	17-Oct-2026
Location / District:	Athani Taluk, Belagavi District, Karnataka	**Analytical Scheme:**	ICAR-NBSS&LUP; Standard Protocol
Soil Classification:	Deep Black Clay / Vertisol	**Target Agro-Zone:**	Karnataka - Belagavi Region

SOIL CHEMICAL AND PHYSICAL PARAMETERS

Parameter Name	Symbol	Observed Value	Unit	Reference Range	Rating / Interpretation
Available Nitrogen	**N**	**315.0**	kg/ha	280.0 - 560.0	Medium-High (Adequate)
Available Phosphorus	**P**	**58.5**	kg/ha	23.0 - 56.0	High (Abundant)
Available Potassium	**K**	**385.0**	kg/ha	140.0 - 330.0	High (Rich)
Soil Reaction (pH)	**pH**	**7.90**	pH scale	6.5 - 7.5	Moderately Alkaline
Electrical Conductivity	**EC**	**0.82**	dS/m	< 1.00	Normal (Approaching Critical)
Organic Carbon	**OC**	**0.78**	%	0.50 - 0.75	High (Rich)

STANDARD ANALYTICAL READOUT (EXTRACTABLE KEY-VALUE STREAM):

Available Nitrogen (N): 315.0 kg/ha [Medium-High (Adequate)]

Available Phosphorus (P): 58.5 kg/ha [High (Abundant)]

Available Potassium (K): 385.0 kg/ha [High (Rich)]

Soil Reaction (pH): 7.90 [Moderately Alkaline]

Electrical Conductivity (EC): 0.82 dS/m [Normal (Approaching Critical)]

Organic Carbon (OC): 0.78 % [High (Rich)]

SOIL HEALTH INTERPRETATION & NUTRIENT ADVISORY

Diagnostic Assessment: Heavy deep black vertisol with strong moisture retention and clay fraction. Rich in potassium and phosphorus reserves, but slight alkaline tendency requires monitoring of micronutrient availability.

General Nutrient Management Guidance: Withhold chemical potassium fertilizer (MOP) to prevent luxury consumption. Utilize acidifying nitrogenous fertilizers such as Ammonium Sulphate. Maintain surface furrow drainage to avoid secondary salt buildup. Favorable for cotton, soybean, and sugarcane.

Soil Analyst: Demonstration Technical Officer

Quality Authorized by: Senior Agricultural Chemist

Status: Fictional Demonstration Data

ACADEMIC DEMONSTRATION DISCLAIMER: *This soil test report contains fictional demonstration data generated strictly for research, user interface validation, and OCR testing in the MannuMitra AI-Based Predictive Farming System. It is not an official laboratory certificate, does not represent physical soil sampling of any real farming parcel, and must not be used as official government documentation or for commercial agricultural transactions.*